Decoupling smoothness, accuracy, and kinematic invariance in biological reach: an ablation study of an equilibrium-point controller in a 34-muscle arm model

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Abstract

Engineering controllers solve musculoskeletal reaching but typically violate the kinematic invariants of human reach: bell-shaped speed profiles, near-straight paths, and a peak-velocity time at 40–50 % of movement duration. For the MyoSuite myoArm (20-DoF, 34 Hill-type muscles) we implement a biologically motivated controller combining (i) Feldman's λ-equilibrium-point hypothesis, (ii) a minimum-jerk virtual trajectory $\boldsymbol{\lambda}(t)$, (iii) a 200 ms visuomotor correction, and (iv) γ-compatible spinal reflexes (Ia, Ib, reciprocal inhibition). Across $n = 50$ randomised targets the full controller is practically equivalent to an **endpoint-PD + spinal** baseline (Cartesian PD descending command paired with the same spinal reflex layer; see §2.6) on minimum tip error (Cohen's $d = +0.03$; paired Wilcoxon detects only a $+10.6$ mm residual against a ≈ 100 mm absolute error, well within a pre-defined ± 20 mm equivalence margin) while halving peak speed (1.78 vs 3.90 m s^{-1}, $d = -7.39$, $p < 10^{-15}$) and reducing jerk by 40 % ($d = -1.74$). Only the variant with stretch reflexes brings the velocity-peak ratio into the canonical human range (0.40–0.50). Straightness stays below the human reference, so we frame the result as a partial reproduction of the bell-shape and smoothness invariants, not full human-like reach. A factorial ablation ($n = 20$) decomposes the contributions: virtual trajectory primarily controls smoothness, visuomotor feedback primarily controls accuracy, and reflexes primarily control velocity-peak timing, with two quantifiable secondary effects reported explicitly. An attempted online cerebellar correction in joint or λ space did not improve performance, consistent with — but not by itself demonstrating — the cerebellum as a slow inverse-model learner rather than a within-trial steering controller. We release a `deterministic_reset` patch for a seeding bug in the MyoSuite reach environments (in the versions tested). The result is mechanistic rather than task-optimal: it attributes separable kinematic axes to distinct biological control layers in a 34-muscle arm.

Keywords: equilibrium-point hypothesis · virtual trajectory · visuomotor feedback · spinal reflexes · musculoskeletal model · reach control · MyoSuite · ablation study

1 Introduction

1.1 Biological reach has structured invariants that engineering controllers tend to violate

Human upper-limb reach exhibits a small set of kinematic regularities that hold across speeds, distances, postures, and even loaded conditions. End-effector speed follows an approximately bell-shaped profile whose peak occurs at 40–50 % of movement duration (Morasso 1981; Flash and Hogan 1985); the hand path between two points is nearly straight in extrinsic space; and the time-derivative of acceleration (jerk) is minimised in the sense of Hogan (1984). These properties are maintained even when the underlying joint trajectories are highly non-linear, suggesting that the central nervous system plans in a representation in which they are natural constraints rather than emergent costs.

Modern musculoskeletal benchmarks such as MyoSuite (Caggiano et al. 2022) make it possible to simulate a 34-muscle Hill-type arm and to test motor-control hypotheses in a setting that retains realistic muscle dynamics. The default control schemes available in this setting — proportional–derivative (PD) endpoint controllers, model predictive control, and reinforcement-learning policies — typically solve the task but produce motion that visibly violates the human invariants: high initial peak speeds (3–4 m s^{-1}), early peak-velocity times (often within 5–10 % of movement duration), and oscillatory residuals near the target. The question we address in this paper is whether a controller that is itself built from biological control hypotheses can match the engineering baseline on accuracy while reproducing the canonical invariants, and if so, which component is responsible for each invariant.

1.2 Theoretical framework: λ-EP, virtual trajectory, and visuomotor feedback

Three mature ideas in motor neuroscience together suggest a complete controller. **Feldman's λ-equilibrium-point hypothesis** (Feldman 1966, 1986; Bizzi et al. 1976) posits that the descending command specifies, for each muscle i, a length threshold λ_i above which the muscle is recruited. Activation is generated locally by $a_i = \text{clip}(c \cdot \max(L_i - \lambda_i, 0),\, 0,\, 1)$ where L_i is the current actuator length; the limb settles into the equilibrium that balances the resulting muscle forces and external loads, with no need for the brain to invert the dynamics. The **virtual-trajectory** extension (Feldman and Levin 1995; Latash 1993) replaces the static $\boldsymbol{\lambda}_{\text{target}}$ with a smooth time-varying trajectory $\boldsymbol{\lambda}(t)$ — for example, a minimum-jerk interpolation — between the initial threshold vector and the target one. Finally, **online visuomotor feedback** (Saunders and Knill 2003; Sarlegna and Sainburg 2009) provides a slow (≈ 100– 200 ms latency) correction in which the central nervous system re-estimates the target threshold from the observed tip position; the error in $\boldsymbol{\lambda}_{\text{target}}$ is corrected without disrupting the ongoing virtual trajectory.

Embedded within the spinal cord, **γ-compatible stretch reflexes** (Prochazka 1989; Loeb 1984), Golgi-tendon load feedback, and reciprocal inhibition shape the activation locally on a 20–30 ms timescale and prevent co-contraction. We retain the architectural slot for descending γ-motoneuron modulation (Prochazka 1989) but disable it in our production controller, since pilot experiments showed no benefit on the metrics of interest (§3, ablation).

These ideas have been tested individually in low-dimensional models or in single-joint preparations. To our knowledge, they have not been combined and ablated factorially in a high-dimensional Hill-type musculoskeletal arm against an engineering baseline.

1.3 Contributions

We make four contributions.

1. We implement a full biological reach controller for MyoSuite that simultaneously incorporates the λ-EP hypothesis, a minimum-jerk virtual trajectory, a periodic visuomotor correction, and a stack of spinal reflexes (Fig. 5). The implementation is MIT-licensed and is archived as Zenodo `10.5281/zenodo.19948021` (GitHub mirror: `jkoba0512/myoarm-lambda-ep`), with reproduction commands documented in the repository README.
2. We show that this controller is practically equivalent to an **endpoint-PD + spinal** baseline — a Cartesian PD descending command sharing the same spinal reflex layer (§2.6) — on minimum tip error ($n = 50$, Cohen's $d = +0.03$; only a small ≈ 10 mm paired residual, within a pre-defined ± 20 mm equivalence margin) while halving peak speed, reducing jerk by 40 %, and bringing the velocity-peak ratio into the canonical human range (Fig. 1). End-effector paths are also straighter than the engineering baseline (Fig. 3), although both controllers fall short of the human straightness reference and the absolute residual error remains ≈ 100 mm; we therefore frame this as a partial — not a full — reproduction of human reach kinematics.
3. We perform a factorial ablation ($n = 20$ per condition) of the three biological components and report a structured decomposition of their effects (Fig. 4): the virtual trajectory dominates the smoothness axis, visuomotor feedback dominates the accuracy axis, and stretch reflexes dominate the velocity-peak-timing axis. We make the contribution structure explicit by reporting both the primary (block-diagonal) and the secondary effects, the latter including a constructive contribution of the virtual trajectory to peak timing and a moderate cost of reflexes on peak-speed regulation.
4. We report a negative result: an online cerebellar correction trained as a forward model and applied either in joint space or in $\boldsymbol{\lambda}$ space does not improve performance in this architecture. We interpret this as consistent with — but not by itself demonstrating — the cerebellum operating as a slow inverse-model learner rather than a within-trial steering controller.

In addition, we identify and patch a seeding bug in the MyoSuite reach environ-

ments, in which `BaseV0.reset(seed=N)` fails to re-seed the internal `np_random` generator and consequently leaks seed-dependent target drift across episodes. We release a small helper, `deterministic_reset`, that restores reproducibility and that we encourage other users of these environments to adopt.

1.4 Paper organisation

Section 2 describes the environment, the controller architecture, and the evaluation protocol. Section 3 reports the headline comparison against the engineering baseline, the factorial ablation, and the negative results on cerebellar correction. Section 4 discusses the biological interpretation, the limitations of the single-environment, single-task evaluation, and the directions in which the framework can be extended.

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2 Methods

2.1 Musculoskeletal environment

We simulate reach in MyoSuite's `myoArmReachRandom-v0` environment (Caggiano et al. 2022; tested with MyoSuite 2.12.1, MuJoCo 3.6.0, Gymnasium 1.2.3 on Linux 6.8), which wraps a MuJoCo (Todorov et al. 2012) model of a human upper limb with 20 kinematic degrees of freedom (shoulder, elbow, forearm, wrist, and the metacarpo-phalangeal / inter-phalangeal joints of the index finger) actuated by 34 Hill-type muscle–tendon units. The end-effector site `IFtip` is the index fingertip. At each episode reset the environment samples a target position uniformly inside a 3-D box around the resting tip; the task is to drive the tip to the target within an episode budget of 12 s (600 control steps at 20 ms per step). The MuJoCo integration step is 2 ms and the controller is queried once per `frame_skip = 10` integration steps, so the effective control period is $\Delta t = 20$ ms. We use the proprioceptive observation $(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{20}$, the actuator length $\mathbf{L} \in \mathbb{R}^{34}$ and velocity $\dot{\mathbf{L}}$, the actuator force $\mathbf{F}_{\text{musc}} \in \mathbb{R}^{34}$, and the tip position $\mathbf{x}_{\text{tip}} \in \mathbb{R}^{3}$ with the reach-error vector $\mathbf{e} = \mathbf{x}_{\text{target}} - \mathbf{x}_{\text{tip}}$.

Seed reproducibility patch (in the MyoSuite versions tested). In the versions of MyoSuite we tested (2.12.x with MuJoCo 3.6.x, Gymnasium 1.2.x), `myosuite/envs/myo/myobase/reach_v0.py` resamples the target inside `BaseV0.reset` before propagating the `seed` argument to the gym RNG, and the parent class only forwards seeds to its embedded robot rather than to the per-environment `np_random` used by the target sampler. Consequently `env.reset(seed=N)` does not, on its own, restore reproducibility in this code path: the same `seed=N` returns different targets across calls because the sampler's RNG state advances between episodes. We patch this with a small helper, `deterministic_reset(env, seed)`, that overwrites `env.unwrapped.np_random` with a freshly seeded generator obtained via `myosuite.utils.seed_envs(seed)` before invoking `env.reset()`. We verified determinism by reseeding the same seed multiple times across interleaved episodes

and confirming that all reset calls returned identical target positions; a minimal reproducer is included in the released repository (`README.md`). All experiments reported below use `deterministic_reset`; we recommend its use whenever reproducible per-seed evaluation is needed in the affected MyoSuite versions, and we encourage users to verify the behaviour in their own version before relying on it.

2.2 λ-equilibrium-point controller

Our controller follows the λ-version of Feldman's equilibrium-point hypothesis (Feldman 1986; Feldman and Levin 1995). The descending command is a vector of muscle threshold lengths $\boldsymbol{\lambda} \in \mathbb{R}^{34}$ rather than a torque or muscle activation. Local muscle activation is generated by the linear stretch above threshold,

$$a_i^{\text{base}} \;=\; \text{clip}\big(c_\lambda \cdot \max(L_i - \lambda_i,\, 0),\; 0,\; 1\big), \qquad i = 1, \dots, 34, \tag{1}$$

with $c_\lambda = 20\,\text{m}^{-1}$ chosen so that a 50 mm stretch saturates a single fiber. We use no inverse dynamics, no torque controller, and no muscle-level Jacobian projection in this branch of the controller — Eq. (1) is the entire descending-to-spinal mapping.

Notation. Throughout, $\mathbf{1}_n \in \mathbb{R}^n$ denotes the all-ones vector of dimension n (distinct from the indicator $\mathbf{1}[\cdot]$ used in §2.4); a scalar added to or subtracted from a vector is shorthand for the corresponding multiple of $\mathbf{1}_n$ of matching dimension. The operations $\text{clip}(\cdot, 0, 1)$ and $\max(\cdot, 0)$ on vector arguments are applied componentwise.

Inverse kinematics for $\boldsymbol{\lambda}_{\text{target}}$. Given a target tip position $\mathbf{x}_{\text{target}}$, we obtain the corresponding equilibrium configuration $\mathbf{q}^*$ by a damped least-squares iteration on the endpoint Jacobian $\mathbf{J}_p = \partial \mathbf{x}_{\text{tip}} / \partial \mathbf{q}$:

$$\Delta \mathbf{q} \;=\; \mathbf{J}_p^\top \big(\mathbf{J}_p \mathbf{J}_p^\top + \mu^2 \mathbf{I}\big)^{-1} (\mathbf{x}_{\text{target}} - \mathbf{x}_{\text{tip}}),$$

with damping $\mu = 0.01$ and a maximum of 30 iterations, with each iterate clipped to the joint range when present. The actuator lengths at the resulting configuration give $\mathbf{L}_{\text{target}} = \mathbf{L}(\mathbf{q}^*)$ via a single forward kinematics pass, and the threshold vector is

$$\boldsymbol{\lambda}_{\text{target}} \;=\; \mathbf{L}_{\text{target}} - \lambda_0 \mathbf{1}_{34}, \tag{2}$$

with a fixed offset $\lambda_0 = 5$ mm so that the equilibrium remains slightly stretched and Eq. (1) does not generate a flat-zero activation at the target. This IK is run once per movement.

Virtual trajectory. Following Feldman and Levin (1995), the descending command is a time-varying trajectory $\boldsymbol{\lambda}(t)$ rather than a static $\boldsymbol{\lambda}_{\text{target}}$. We use a min-jerk interpolation (Flash and Hogan 1985) between the threshold at movement onset $\boldsymbol{\lambda}_s = \mathbf{L}(\mathbf{q}_0) - \lambda_0 \mathbf{1}_{34}$ and the target threshold:

$$\boldsymbol{\lambda}(\tau) \;=\; \boldsymbol{\lambda}_s + s(\tau)\big(\boldsymbol{\lambda}_{\text{target}} - \boldsymbol{\lambda}_s\big), \qquad s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5, \tag{3}$$

with normalised time $\tau = t/T \in [0, 1]$. The movement duration is selected adaptively from the target distance, $T = \mathrm{clip}\big(\|\mathbf{x}_{\text{target}} - \mathbf{x}_{\text{tip}}\| \cdot 1.2/0.5,\ 0.3,\ 2.5\big)$ seconds, which corresponds to a nominal 0.5 m s^{-1} movement speed scaled by 1.2× and bounded for short and long reaches respectively. The activation in Eq. (1) is computed using $\boldsymbol{\lambda}(t)$ in place of $\boldsymbol{\lambda}_{\text{target}}$.

2.3 Online visuomotor feedback

The static virtual trajectory of Eq. (3) is open-loop in the visual sense: it commits to the threshold trajectory implied by the IK solution at movement onset and is blind to the actual tip motion. Saunders and Knill (2003) and Sarlegna and Sainburg (2009) show that human reach incorporates online visual updates with a 100–200 ms latency. We add an analogous slow loop that re-runs the IK from the current configuration and updates only the target threshold,

$$\boldsymbol{\lambda}_{\text{target}} \leftarrow \mathbf{L}\big(\mathrm{IK}(\mathbf{x}_{\text{target}}; \mathbf{q})\big) - \lambda_0 \mathbf{1}_{34}, \qquad \text{every } P_{\text{vm}} \Delta t = 200\,\text{ms}. \tag{4}$$

We deliberately do not reset $\boldsymbol{\lambda}_s$ or τ: resetting either pulls $s(\tau)$ back to zero, which makes $\boldsymbol{\lambda}(t)$ collapse onto the current configuration and zeros out the descending drive in Eq. (1) — empirically the limb stalls. Updating only the endpoint of the trajectory preserves both continuity of $\boldsymbol{\lambda}(t)$ and the temporal scaffold of the min-jerk profile. The update period $P_{\text{vm}} = 10$ control steps corresponds to 200 ms, near the upper bound of the human range.

2.4 Spinal reflexes

Three components shape the activation locally on a faster timescale than the visuomotor loop:

Ia stretch reflex. Lengthening muscle spindles produce a velocity-proportional excitation $\Delta a_i^{\text{Ia}} = K_{\text{Ia}} \cdot \max(\dot{L}_i,\, 0)$, with $K_{\text{Ia}} = 0.05$ s and a proprioceptive transmission delay of 20 ms (10 integration steps), implemented with a fixed-length delay buffer. The architecture admits a γ-motoneuron modulation $K_{\text{Ia}} \to K_{\text{Ia}}\,(1 + \gamma \exp(-\|\mathbf{e}\|/\sigma_\gamma))$ on target approach (Prochazka 1989), but $\gamma = 0$ in all configurations reported below (γ-on configurations were tested in pilot experiments and provided no benefit).

Ib Golgi-tendon reflex. Muscle force above a threshold produces inhibitory feedback $\Delta a_i^{\text{Ib}} = -K_{\text{Ib}} \cdot \max(|F_{\text{musc,i}}| - F_{\text{Ib}},\, 0)$, with $K_{\text{Ib}} = 0.03\,\text{N}^{-1}$ and $F_{\text{Ib}} = 200$ N.

Reciprocal inhibition. Antagonist pairs are inhibited when agonist activation exceeds a threshold, $\Delta a_i^{\text{RI}} = -K_{\text{RI}} \cdot \mathbf{1}[a_{j(i)} > \theta_{\text{RI}}] \cdot a_{j(i)}$, where $j(i)$ is the antagonist of muscle i in the MyoSuite agonist–antagonist tabulation, with $K_{\text{RI}} = 0.5$ and $\theta_{\text{RI}} = 0.3$.

The total activation sent to the muscle plant is the saturated sum

$$a_i^{\text{total}} \;=\; \text{clip}\big(a_i^{\text{base}} + \Delta a_i^{\text{Ia}} + \Delta a_i^{\text{Ib}} + \Delta a_i^{\text{RI}} + \Delta a_i^{\text{cereb}},\, 0,\, 1\big), \tag{5}$$

where the cerebellar correction $\Delta\mathbf{a}^{\text{cereb}}$ is described in §2.5 and is zero in the configurations used for the headline comparison and ablation.

2.5 Cerebellar forward model (negative-result branch)

To test whether an online cerebellar correction can further improve performance, we trained a closed-form-continuous-time (CfC) recurrent forward model (Hasani et al. 2022) that predicts the next-step joint configuration from the delayed proprioceptive state and an efference copy of the descending command: $\hat{\mathbf{q}}_{t+1} = f_{\text{CfC}}(\mathbf{q}_t^{\text{del}}, \dot{\mathbf{q}}_t^{\text{del}}, \mathbf{a}_t^{\text{efcopy}})$. We evaluated two correction pathways inspired by competing interpretations of cerebellar output (Kawato 1999; Wolpert et al. 1998; Ito 2008). The joint-space pathway maps the prediction error $\delta\mathbf{q} = \mathbf{q}_{t+1} - \hat{\mathbf{q}}_{t+1}$ to a torque correction $\boldsymbol{\tau}_{\text{cereb}} = K_{\text{cereb}}\,\delta\mathbf{q}$ and projects it into muscle space through the actuator-Jacobian pseudoinverse $\Delta\mathbf{a}^{\text{cereb}} = \mathbf{J}_{\text{act}}^{+}\boldsymbol{\tau}_{\text{cereb}}$. The λ-space pathway, more consistent with the EP framework, maps $\delta\mathbf{q}$ directly into a threshold correction $\Delta\boldsymbol{\lambda} = -K_{\text{cereb}}^{\lambda}\,\mathbf{R}\,\delta\mathbf{q}$, where $\mathbf{R} = \partial\mathbf{L}/\partial\mathbf{q}$ is the muscle moment-arm matrix; this correction adds to $\boldsymbol{\lambda}(t)$ before Eq. (1). Both pathways are passed through a 30 ms cerebello-cortical loop delay. Online learning of the CfC weights is gated by an inferior-olive analogue that fires sparsely on prediction-error spikes (Ito 2008); the firing-rate calibration and module are documented with the released code. In the headline ablation we report two cerebellar configurations with $K_{\text{cereb}} = 0.2$ (joint pathway) and $K_{\text{cereb}}^{\lambda} = 0.5$ (λ pathway) on top of the full λ-EP controller with reflexes; both gains were the best among a small sweep ($K_{\text{cereb}} \in \{0.05, 0.1, 0.2, 0.5\}$, $K_{\text{cereb}}^{\lambda} \in \{0.1, 0.5, 1.0\}$).

2.6 Endpoint-PD baseline

For the comparison condition we use a conventional endpoint PD-with-integral controller. A virtual force is computed in Cartesian space,

$$\mathbf{F}_{\text{ee}} \;=\; K_p\mathbf{e} - K_d\dot{\mathbf{x}}_{\text{tip}} + K_i\int \mathbf{e}\,\mathrm{d}t, \tag{6}$$

with $K_p = 80$ N m^{-1}, $K_d = 15$ N s m^{-1}, $K_i = 2$ N m^{-1} s^{-1} and the integral term clipped to ± 2 m s. The force is mapped to joint torques by $\boldsymbol{\tau}_{\text{pd}} = \mathbf{J}_p^{\top}\mathbf{F}_{\text{ee}}$, gravity-compensated by adding the bias-force vector $\boldsymbol{\tau}_{\text{grav}}$ returned by MuJoCo's `mj_data.qfrc_bias`, and projected into muscle space through the actuator-Jacobian pseudoinverse with a small static activation bias, $\mathbf{a} = \text{clip}(\mathbf{J}_{\text{act}}^{+}(\boldsymbol{\tau}_{\text{pd}} + \boldsymbol{\tau}_{\text{grav}}) + a_{\text{bias}}\mathbf{1}_{34},\, 0,\, 1)$, $a_{\text{bias}} = 0.15$. The actuator Jacobian is recomputed every 50 steps by central finite differences. We refer to this baseline as **endpoint-PD + spinal** throughout: it shares the spinal reflex layer (Ia, Ib, RI; §2.4) with the full λ-EP controller, so that the headline comparison contrasts the descending command

(Cartesian endpoint PD vs λ-EP with virtual trajectory and visuomotor feedback) rather than spinal feedback.

A small implementation asymmetry exists in the originally registered PD baseline: the joint-pathway cerebellar correction (§2.5, $K_{\text{cereb}} = 0.2$, pre-trained CfC checkpoint) is enabled in that configuration, whereas the full λ-EP controller has no cerebellar correction. We ran a no-cerebellum control (F17, identical seed list, $n = 50$) and confirmed that the cerebellar branch acts as a no-op on this baseline (median paired differences ≤ 0.6 mm on minimum tip error, exactly zero on peak speed and vpr; complete metric × test table provided in the released supplementary results). The asymmetry therefore does not confound the headline comparison; the factorial ablation in §3.2 disentangles the spinal contribution separately.

2.7 Evaluation protocol

Test seeds. For each condition we evaluate on $n = 50$ randomised target seeds. We draw the seed pool from the integers $0, 1, \ldots, 149$ and retain the first 50 for which the sampled target is reachable, defined by $\|\mathbf{e}_0\| < 0.85$ m at reset; this threshold is conservative relative to the model's full workspace and filters only the long-tail of unreachable samples (e.g. far behind the shoulder). The selected seeds are stored alongside the results for reproducibility. For the factorial ablation (Fig. 4) we use $n = 20$ seeds drawn from `range(50)` to match earlier hypothesis-test runs; for the headline comparison and the velocity- peak-ratio distribution (Figs 1–3) we use the full $n = 50$.

Episode protocol. We run 600 control steps (12 s) per seed. At every step we record $\mathbf{x}_{\text{tip}}$ and $\|\mathbf{e}\|$, and at the end of the episode we extract the movement window as the contiguous interval where the tip speed exceeds 0.02 m s^{-1}, with onset and offset detected forward and backward from the speed maximum.

Metrics. Within the movement window we compute the following quantities, all from $\mathbf{x}_{\text{tip}}$ alone: minimum tip error $\min_t \|\mathbf{e}(t)\|$ in mm; final tip error $\|\mathbf{e}(T_{\text{end}})\|$; direction error in degrees, $\arccos(\hat{\mathbf{u}}_{\text{travel}} \cdot \hat{\mathbf{u}}_{\text{target}})$ between the net travel and the target direction; progress ratio, the projection of net travel onto the target direction normalised by target distance; peak speed $\max_t \|\dot{\mathbf{x}}_{\text{tip}}\|$; velocity-peak ratio (vpr), the time of peak speed normalised by movement duration (human reference: 0.40–0.50; Morasso 1981; Flash and Hogan 1985); root-mean-square jerk on the third time-derivative of $\mathbf{x}_{\text{tip}}$; straightness, the ratio of the straight-line distance between movement onset and offset to the arc-length of the path (1 = perfectly straight); and speed-profile skewness (third standardised moment) within the movement window, used only as a secondary descriptor of bell shape.

Statistics. We use two distinct reference conditions for the two analyses:

1. Headline comparison (Table 1, Figs 1–3, §3.1). The reference is the **endpoint-PD + spinal baseline** described in §2.6 (M1-level contrast, with the spinal layer held common).

2. Factorial ablation (Fig. 4, §3.2). The reference is the **pure λ + visuomotor controller** (no reflexes, no cerebellar correction).

Because deterministic_reset guarantees that the same seed produces the same target across conditions, the per-seed metrics are paired by design. Our **primary** inferential test is therefore the two-sided **paired Wilcoxon signed-rank test** on the per-seed value pairs. We also report Welch's two-sided t-test (unequal variances) as a confirmatory analysis and a vehicle for the distribution-level effect size, and we report **Cohen's** $d = (\mu_a - \mu_b)/\sigma_p$, $\sigma_p = \sqrt{(\sigma_a^2 + \sigma_b^2)/2}$, as the primary effect-size measure throughout. For Fig. 4, d is signed so that positive = improvement (lower for error / peak speed / jerk; higher for vpr and progress ratio; vpr is folded toward the human band by using $|0.45 - \text{vpr}|$ before signing). Significance markers are $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$ and refer to the paired test unless stated otherwise.

Pre-defined equivalence margins for accuracy. To support the practically-equivalent claim against the headline baseline (§3.1) we define the following practical equivalence margins on the per-seed paired differences and run a two one-sided t-test (TOST, Lakens 2017):

- Minimum tip error: ± 20 mm, chosen as one-fifth of the absolute residual (≈ 100 mm) carried by both controllers.
- Final tip error: ± 25 mm, chosen on the same scale.
- Direction error: $\pm 5°$, chosen as one-fifth of the typical unconstrained-vision human reach direction error.

Equivalence is established if both lower- and upper-bound paired t-tests reject at $\alpha = 0.05$. We do not define equivalence margins for peak speed, jerk, vpr, or straightness because the effect sizes there are large and the controllers are not expected to be equivalent on those axes — the contrast is the point.

Compute and reproducibility. Each condition × seed run takes ≈ 6 s on a single CPU; the headline $n = 50$ comparison across six conditions completes in under 35 minutes. The MIT- licensed implementation, exact seed lists, raw per-seed metric tables, and trained CfC weights are archived as Zenodo `10.5281/zenodo.19948021` (GitHub mirror: `jkoba0512/myoarm-lambda-ep`, release `v1.0.0-bioRxiv`), with reproduction commands documented in the repository `README`.

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3 Results

3.1 The full controller matches PD accuracy with halved peak speed and human-range peak timing

Across $n = 50$ randomised reach targets, the full λ-EP controller (virtual trajectory + visuomotor feedback + spinal reflexes; no cerebellar correction) and the **endpoint-PD + spinal** baseline produce endpoint accuracy that is practically

equivalent but not statistically identical. Minimum tip error is similar for the two controllers (103.7 ± 58.2 mm for the baseline vs 105.3 ± 66.8 mm for the full controller; Cohen's $d = +0.03$). A two one-sided t-test (TOST) on the per-seed paired differences with a pre-defined ± 20 mm equivalence margin **rejects both equivalence-null tests** (lower-bound $p = 7 \times 10^{-4}$, upper-bound $p = 0.003$, $p_{\text{TOST}} = 0.003$), so the paired mean difference lies inside the ± 20 mm equivalence interval and the two controllers are statistically equivalent on minimum tip error within that margin. The same TOST analysis with the margins pre-defined in §2.7 also establishes equivalence on final tip error (± 25 mm; $p_{\text{TOST}} = 1.8 \times 10^{-4}$) and on direction error ($\pm 5°$; $p_{\text{TOST}} = 1.1 \times 10^{-8}$). The seed-paired Wilcoxon test nevertheless detects small residual shifts on minimum tip error ($p = 0.047$, median $+10.6$ mm, $d = +0.03$; Table 1, Fig. 1d) and on direction error ($p = 0.039$, median $+2.0°$, $d = +0.19$), both inside the practical equivalence margins. Final tip error is paired-equivalent (135.4 vs 129.7 mm, paired $p = 0.78$, $d = -0.10$) and progress ratio does not differ ($d \leq 0.32$). We read these two families of tests jointly: the full controller is bounded-equivalent to the baseline within the practical margins, with small detectable shifts inside those margins. We treat the residual accuracy effects as practically negligible relative to the ≈ 100 mm absolute residual that both controllers carry, and we discuss the distinction between practical and statistical equivalence in §4.4. Solve rate is low for both controllers (1/50 vs 2/50) because the MyoSuite reach environment uses a tight 50 mm tolerance, smaller than the residual error reported in unconstrained human reaching without a final corrective movement (Sarlegna and Sainburg 2009); we report all metrics on the full $n = 50$ regardless of solve status.

Smoothness. The biological controller more than halves peak tip speed relative to the engineering baseline (1.78 ± 0.26 vs 3.90 ± 0.31 m s^{-1}; paired Wilcoxon $p < 10^{-15}$, Cohen's $d = -7.39$; Welch confirms with $t = -36.97$, $p < 10^{-25}$; Fig. 1a) and reduces root-mean-square jerk by ≈ 40 % (240 ± 70 vs 399 ± 109 m s^{-3}; paired $p < 10^{-10}$, $d = -1.74$; Welch $p < 10^{-12}$; Fig. 1b). The 1.78 m s^{-1} peak speed is consistent with reported peak hand speeds for unconstrained 30–50 cm reaches (Morasso 1981).

Peak timing inside the human band; residual asymmetry. Only the full controller produces a velocity-peak ratio (vpr) inside the human reference range. Mean vpr is 0.064 ± 0.030 for the baseline (the speed peak occurs in the first 6 % of the movement window, reflecting the near-impulsive ballistic profile of feedback-linearisation control), 0.221 ± 0.170 for the pure λ + visuomotor controller without reflexes (a broad, double-peaked profile), and 0.403 ± 0.122 for the full controller with reflexes — squarely inside the 0.40–0.50 human band (Flash and Hogan 1985). Speed-profile skewness drops from $+2.56$ for the baseline to $+1.10$ for the full controller ($d = -2.69$, $p < 10^{-22}$), indicating a markedly more symmetric profile. We emphasise that peak timing is what enters the human band; the full speed profile is not a perfectly symmetric bell — the residual right-skewness ($+1.10$) and the small initial transient visible in Fig. 2a (≈ 1.06 m s^{-1} at $t = 0.05$ s, discussed below) remain. We therefore claim a partial

reproduction of the bell-shape invariant: peak timing in the human range, with residual profile asymmetry.

Path geometry. End-effector paths from the full controller are also straighter than those from endpoint-PD (0.612 vs 0.531, $d = +0.75$, $p < 10^{-3}$; Fig. 3a–b). The endpoint-PD paths overshoot the target along the initial ballistic direction and then oscillate around the residual offset; the λ-EP paths approach the target on a smoother arc and the distance-to-target curve is monotone over the movement window (Fig. 3c).

Figure 2 shows the per-seed speed profiles in three panels. For a representative seed (seed 27, vpr $= 0.453$, in the human range), the endpoint-PD controller produces an initial spike at ≈ 3.9 m s^{-1} followed by oscillatory residuals (Fig. 2a, gray), the pure λ + visuomotor controller (Fig. 2a, light blue) shows a broad double-peaked profile, and the +reflex controller produces a single dominant bell near 1.7 m s^{-1} peaking at 45 % of the movement window (Fig. 2a, dark blue). Time-warping the per-seed profiles to a common normalised duration and averaging across seeds shows that only the +reflex condition has a mean profile peaking inside the human band (Fig. 2b). The per-seed vpr distribution (Fig. 2c) is centred inside the human band only for the +reflex condition.

We retain a small initial transient (≈ 1.06 m s^{-1} at $t = 0.05$ s) at the onset of the +reflex profile in Fig. 2a. This is not a smoothing artefact: it reflects the impulsive response of the muscle plant to the step in $\boldsymbol{\lambda}_{\text{target}}$ at movement onset (the initial-posture-dependent component of the response is identical across seeds), and it is consistent with the residual right skewness of the speed profile (+1.10). We discuss this in §4.

Table 1: Headline comparison ($n = 50$). Mean ± SD; Cohen's d is reported as the distribution-level effect size and the asterisk column reports the result of the seed-paired Wilcoxon signed-rank test ($^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, n.s. = not significant) — see §2.7. The asterisks refer to the paired test, not to the magnitude of d. The full controller is the λ-EP controller with virtual trajectory, visuomotor feedback, and spinal reflexes (no cerebellar correction); the baseline is endpoint-PD with the same spinal reflex layer.

Metric	endpoint-PD + spinal	full controller	Cohen's d	paired p
Min. tip error [mm]	103.7 ± 58.2	105.3 ± 66.8	$+0.03$	0.047^{*}
Final tip error [mm]	135.4 ± 55.5	129.7 ± 62.2	-0.10	0.78 n.s.
Direction error [°]	7.2 ± 4.8	8.1 ± 4.7	$+0.19$	0.039^{*}
Peak speed [m s^{-1}]	3.90 ± 0.31	1.78 ± 0.26	-7.39	$< 10^{-15}$ ***
Jerk-rms [m s^{-3}]	399 ± 109	240 ± 70	-1.74	$< 10^{-10}$ ***
Velocity-peak ratio	0.064 ± 0.030	$\mathbf{0.403 \pm 0.122}$	$+3.81$	$< 10^{-9}$ ***
Straightness ratio	0.531 ± 0.115	0.612 ± 0.101	$+0.75$	$< 10^{-3}$ ***
Speed-profile skewness	$+2.56$	$+1.10$	-2.69	$< 10^{-22}$ ***

3.2 Each component contributes preferentially to one task axis, with structured cross-talk

To attribute the headline result to its components, we ran a factorial ablation in which we removed (or added) each of the three biological components from the full λ-EP controller and re-evaluated on $n = 20$ targets (Fig. 4). This lower-powered ablation is used for component attribution rather than final performance estimation; the headline performance estimates are reported from the $n = 50$ comparison of §3.1. We treat the pure λ + visuomotor controller (no reflexes, no cerebellar correction) as the reference configuration and report signed Cohen's d in the convention positive = improvement relative to that reference (lower for error / peak speed / jerk; higher for vpr; $|0.45 - \text{vpr}|$ folded toward the human band before signing).

The ablation reveals a primary + secondary structure rather than strict orthogonality:

Virtual trajectory primarily controls smoothness. Replacing the min-jerk virtual trajectory by a static $\boldsymbol{\lambda}_{\text{target}}$ inflates peak speed by an order of magnitude ($1.47 \to 4.32$ m s^{-1}, $d = +3.87$, $p < 10^{-10}$ relative to the reference) and doubles jerk ($d = +2.03$, $p < 10^{-5}$). It has no significant effect on tip error ($d = -0.10$, n.s.) or direction error ($d = +0.02$, n.s.). It does, however, contribute constructively to peak timing: removing it worsens vpr by $d = -1.07$ ($p = 0.003$). The virtual trajectory's contribution is therefore primarily smoothness (peak speed, jerk) with a secondary positive contribution to velocity-peak timing.

Visuomotor feedback primarily controls accuracy. Removing the 200 ms visuomotor update worsens minimum tip error by $d = +0.67$ (paired $p < 10^{-4}$, with a $+34.5$ mm median per-seed shift) and direction error by $d = +0.89$ (Welch $p = 0.008$, broadly consistent under the paired test). It has no paired-significant effect on peak speed, jerk, or vpr ($|d| \leq 0.19$, paired $p \geq 0.08$). Visuomotor feedback also worsens straightness when removed ($d = +1.34$, paired $p < 10^{-3}$) — paths become shorter (less wandering) without the feedback, but at the cost of accuracy. We interpret the visuomotor contribution as accuracy-dominant with a small positive effect on the path length via continuous re-aiming.

Spinal reflexes primarily control velocity-peak timing. Adding reflexes to the reference configuration improves vpr by $d = +1.44$ (paired $p < 10^{-3}$) but worsens peak-speed control by $d = +1.55$ (paired $p < 10^{-5}$) — i.e. the +reflex controller produces a larger peak-speed magnitude than the reflex-free reference, although still about half that of endpoint-PD. The seed-paired Wilcoxon test also detects two small accuracy costs that Welch did not flag at $n = 20$: reflexes increase minimum tip error by a median of 7.5 mm ($d = +0.17$, paired $p = 0.024$) and direction error by a small but paired-significant amount ($d = +0.23$, paired $p < 0.05$). These shifts are about an order of magnitude smaller than the costs on velocity-peak timing and peak speed, and an order of magnitude smaller than the ≈ 100 mm absolute residual; we therefore class them as secondary effects but report them explicitly. The reflex contribution is therefore primarily velocity-peak timing, with

secondary costs on peak-speed control (large) and on accuracy (small). This peak-timing/peak-speed trade-off is the source of the small initial-transient bump visible in Fig. 2a.

The full picture (Fig. 4b) is a near-block-diagonal pattern (virtual trajectory $\rightarrow$ smoothness; visuomotor $\rightarrow$ accuracy; reflexes $\rightarrow$ velocity-peak timing) with two off-diagonal terms of practical size — VT$\rightarrow$peak timing (constructive, $d = +1.07$, paired $p < 0.01$) and reflex$\rightarrow$peak-speed control (cost, $d = -1.55$, paired $p < 10^{-5}$) — and with two small but paired-significant off-diagonal accuracy costs of reflexes ($d = -0.17$ on min tip error, $d = -0.23$ on direction error; both $p < 0.05$). We make this structure explicit in the figure rather than collapsing it into a single "orthogonal" claim, because each off-diagonal term has a mechanistic interpretation that we discuss below (§4). All significance markers in this section refer to the paired Wilcoxon test, with Welch's t-test as a confirmatory analysis (cf. §2.7).

As a sanity check we re-ran the engineering baseline (endpoint-PD with the spinal layer retained, see §2.6) under the ablation seed list and recovered the headline contrast against the same reference configuration: peak speed worsens by $d = +9.54$ ($p < 10^{-24}$) and vpr by $d = -1.38$ ($p < 10^{-3}$), with no significant change in tip error. We do not claim that removing all three components from the λ-EP controller is equivalent to endpoint-PD: the static-λ configuration (− VT, − visuomotor) still drives muscles via Eq. (1) and is conceptually different from a Cartesian PD controller projected through $\mathbf{J}_{\text{act}}^{+}$.

3.3 Online cerebellar correction does not improve performance

Two cerebellar correction pathways were evaluated on top of the full λ-EP+reflex controller on $n = 50$ seeds: (i) a joint-space pathway (CfC $\rightarrow$ torque correction $\rightarrow$ $\mathbf{J}_{\text{act}}^{+}$ projection) at $K_{\text{cereb}} = 0.2$, and (ii) a λ-space pathway (CfC $\rightarrow \Delta\boldsymbol{\lambda} = -K^{\lambda}\,\mathbf{R}\,\delta\mathbf{q}$) at $K_{\text{cereb}}^{\lambda} = 0.5$. Neither improves performance on any metric. Relative to the cerebellum-free full controller, the joint-space correction is essentially a no-op — minimum tip error 105.3 ± 66.6 vs 105.3 ± 66.8 mm, peak speed 1.80 vs 1.78 m s^{-1}, vpr 0.407 vs 0.403 — because the $\mathbf{J}_{\text{act}}^{+}$ projection of the cerebellar torque correction is small after the spinal layer has already integrated proprioceptive feedback. The λ-space correction degrades the bell shape (vpr 0.330 vs 0.403, $d \approx -0.62$) without any compensating accuracy gain. We swept gains in $\{0.05, 0.1, 0.2, 0.5\}$ for the joint pathway and $\{0.1, 0.5, 1.0\}$ for the λ pathway and found no setting that simultaneously improves accuracy, smoothness, and bell shape; the full sweep table is included in the released code archive (see Zenodo record above).

We interpret this negative result not as evidence against a role for the cerebellum in reaching, but as a constraint on its temporal mode of action in this controller architecture. With the spinal reflex layer already correcting proprioceptive errors at 20 ms, an additional 30 ms cerebellar correction is fighting the same error twice; with a fixed pre-trained CfC forward model, the correction provides no information that the spinal layer has not already used. This is consistent with the view that the cerebellar contribution to reaching is dominated by trial-by-trial learning of

internal models rather than within-trial steering (Wolpert et al. 1998; Ito 2008; discussion in §4).

3.4 Validation of the seed-reproducibility patch

We validated the `deterministic_reset` helper described in §2.1 by repeating three trials of an identical reset schedule (`seeds = [0, 1, 2, 0, 7, 0]`, 18 resets in total) and confirming that all calls with the same `seed` returned the same target position to floating-point equality. Without the patch, the same reset schedule returns six different target positions for the three calls with `seed = 0`, because the env-internal `np_random` advances between calls. The patch is a one-line wrapper that re-seeds `env.unwrapped.np_random` with `myosuite.utils.seed_envs(seed)` before calling `env.reset()`; we encourage all users of these reach environments to adopt it for reproducible per-seed evaluation.

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4 Discussion

4.1 Bell-shape and smoothness invariants emerge from biologically motivated control

The headline result of this paper is that an entirely biologically-grounded controller — Feldman's λ-equilibrium-point hypothesis with a min-jerk virtual trajectory, a slow visuomotor loop, and γ-compatible spinal reflexes — is practically equivalent to an endpoint-PD + spinal baseline on minimum tip error (Welch's $d = +0.03$, n.s.; paired Wilcoxon detects a small ≈ 10 mm residual that we discuss in §4.4, well within the pre-defined ± 20 mm equivalence margin) while halving peak speed, reducing jerk by 40 %, and bringing the velocity-peak ratio into the canonical human reference range. The result speaks against the view, sometimes implicit in the engineering motor-control literature, that the bell-shape and smoothness invariants of human reach are an aesthetic "smoothness penalty" added on top of an underlying inverse-dynamics solution. Under the controller architecture studied here, these two invariants are not added on top — they can emerge in this architecture when the λ-equilibrium-point controller is paired with a physiologically plausible spinal layer and a minimum-jerk descending trajectory. We are careful not to overstate this: the architecture has tunable parameters (notably c_λ, K_{Ia}, and the T schedule) and the result is restricted to two of the canonical kinematic invariants. End-effector straightness and the absolute residual error remain limitations that we discuss in §4.4.

4.2 Why these specific component-to-axis assignments

The factorial ablation makes the assignment biologically intuitive. The virtual trajectory is the only component that operates on the full duration of the reach: it imposes a smooth time-course on the descending command itself, and so its removal directly inflates the peak rate of change of muscle activation, which propagates

through the muscle plant to high tip speed and high jerk. Its $d = +1.07$ secondary contribution to peak timing is consistent with the fact that any descending command with a smooth velocity-time profile already starts the limb moving with the correct sign of asymmetry, before reflexes shape the late phase.

The visuomotor loop is a slow estimator of the target in λ space. Its removal does not change the smoothness of the trajectory — the min-jerk profile in λ-space is the same — but it removes the correction that compensates for the discrepancy between the IK solution at movement onset and the actual configuration evolution under muscle dynamics. This discrepancy accumulates as a residual offset that the open-loop trajectory cannot retire, and it shows up exactly where we predict, on minimum tip error and direction error.

The spinal reflex layer is the only component that adds a closed feedback loop on the muscle-length axis at a 20 ms timescale. Its $d = +1.44$ contribution to vpr is precisely the kind of effect predicted by Prochazka's (1989) sensorimotor-gain perspective and the "preflex" concept of Loeb (1984): the antagonist activation triggered by stretch reflex in the late phase of the reach decelerates the limb on the descending side of the bell, which both moves the peak to mid-movement and sharpens the peak. The cost of this on peak-speed regulation ($d = -1.55$) is the price the limb pays for this active deceleration: by the time the reflex acts, the limb is already moving fast, so the overall peak speed remains higher than in the reflex-free configuration. We interpret this as the biological signature of an energy-shaping deceleration controller rather than an energy-minimising one.

4.3 Cerebellum as a slow learner, not a within-trial steerer

The negative cerebellar result is robust across two correction pathways and a small gain sweep. We do not interpret it as evidence against cerebellar involvement in reaching — there is overwhelming evidence for that — but as a constraint on the time scale at which the cerebellum contributes useful information in this controller architecture. With a pre-trained CfC forward model and a 30 ms cerebello-cortical loop delay, the corrections we evaluated are applied late enough that the spinal layer has already retired the prediction error using its own 20 ms loop. The corrections that remain to be made are of the kind that require many trials to estimate stably — for example, slow drifts in muscle properties, inertial loads, or the geometry of the hand–object interaction. Our controller does not yet adapt across trials, which we take as the natural next step. The result is consistent with the cerebellar contribution being primarily inverse-model learning (Wolpert et al. 1998; Kawato 1999; Ito 2008) rather than within-trial steering.

4.4 Limitations

Solve rate. Both the engineering baseline and the biological controller achieve a low solve rate (1–2 / 50) on the standard MyoSuite reach tolerance (50 mm). The two controllers reach to within ≈ 100 mm of the target, so the limiting factor is the joint combination of controller, environment, and tolerance threshold

rather than any single one of them. The 50 mm tolerance is itself tight relative to the residual error reported for unconstrained-vision human reach without a final corrective sub-movement (Sarlegna and Sainburg 2009). We deliberately report metric distributions rather than solve rate as the primary outcome because the present paper evaluates kinematic-invariant attribution across control layers, not task-completion performance; a task-completion-oriented controller would close the final ≈ 50 mm with a corrective sub-movement that we do not model.

Straightness. The full controller's straightness ratio of 0.612 is below the human range of ≈ 0.85 for medium-speed reaches (Morasso 1981). Inspection of the trajectories shows that this is dominated by post-target path wandering during the late phase of the episode (the reach window is 12 s but the movement window is ≈ 2 s), not by curvature during the approach. Restricting the analysis to the approach phase gives a straightness ratio of ≈ 0.78, much closer to the human range. A task-space virtual trajectory (Morasso 1981; Flash and Hogan 1985) computed in end-effector coordinates rather than in λ space did improve approach straightness in pilot experiments (F15) but was unstable when combined with the visuomotor loop, and we did not include it in the production controller.

Single environment, single body site. All results are on the `myoArmReachRandom-v0` environment with the index fingertip as the end-effector. Whether the same component-to-axis assignments hold on the `myoArmReachFixed` environment, on tasks with non-pointing end- effectors, or under loaded conditions is left to future work.

No RL baseline. We compare against the endpoint-PD + spinal baseline but not against state-of-the-art RL controllers such as DEP-RL (Schumacher et al. 2023). Our argument is not that the biological controller beats RL — it almost certainly does not on aggregate task performance — but that the kinematic invariants characteristic of human reach can be reproduced by a controller built from the biological hypotheses without sacrificing accuracy. A direct RL comparison is left to a longer follow-up paper.

Initial transient. The ≈ 1.06 m s^{-1} transient at the onset of the +reflex speed profile (Fig. 2a) is a real artefact of the step in $\boldsymbol{\lambda}_{\text{target}}$ at movement onset, amplified by the muscle plant's near-impulsive response from a neutral resting state. It is consistent with the residual right- skewness of the bell ($+1.10$). A smoother onset shape (e.g. a ramp on $\boldsymbol{\lambda}_{\text{target}}$ during the first 50–100 ms) would likely remove it; we did not include this in the production controller because Eq. (3) already produces a smooth $\boldsymbol{\lambda}(t)$ from $\boldsymbol{\lambda}_s$ and the transient is an artefact of the neutral-posture initial condition, not of the algorithm.

4.5 Implications and future directions

The structured contribution decomposition suggests two operationally useful design principles for biologically-grounded musculoskeletal controllers. First, the smoothness, accuracy, and velocity-peak- timing axes can be tuned semi-

independently, because each is dominated by a different anatomical layer (cortical descending command, slow visuomotor loop, spinal reflex). This makes the controller easier to debug and to extend than an end-to-end policy in which all three are entangled. Second, the cerebellar contribution is not a within-trial controller in this architecture; the natural next step is to make the CfC forward model adapt across trials and to evaluate whether trial-by-trial adaptation can match human force-field and visuomotor-rotation adaptation curves. We expect such an extension to address the limitations on solve rate and on straightness without disturbing the component-to-axis assignment reported here.

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5 Conclusion

A complete biologically-grounded reach controller — Feldman's λ-equilibrium-point hypothesis with a minimum-jerk virtual trajectory, a 200 ms visuomotor correction loop, and a stack of γ-compatible spinal reflexes — is practically equivalent to an endpoint-PD + spinal baseline on minimum tip error in the MyoSuite myoArm (only a ≈ 10 mm paired residual against a ≈ 100 mm absolute error, within a pre-defined ± 20 mm equivalence margin) while halving peak tip speed, reducing jerk by 40 %, and bringing the velocity-peak ratio into the canonical human range. We do not claim full human-like reach: absolute endpoint error remains ≈ 100 mm and straightness is below the human reference, so the result is a partial reproduction of the bell-shape and smoothness invariants rather than a full one. A factorial ablation reveals a structured primary + secondary decomposition of these effects: the virtual trajectory primarily controls smoothness, the visuomotor loop primarily controls accuracy, and stretch reflexes primarily control velocity-peak timing, with two practically-sized off-diagonal contributions that we report explicitly. An online cerebellar correction in either joint or λ space does not improve performance — consistent with, but not by itself demonstrating, the cerebellum operating as a slow inverse-model learner rather than a within-trial steering controller. We release a `deterministic_reset` helper that fixes a seed-reproducibility bug in the MyoSuite reach environments (in the versions tested). The controller should not be read as a task-optimal reaching policy; it is a mechanistic ablation showing how distinct biological control layers shape separable kinematic axes in a high-dimensional muscle arm.

Figure captions

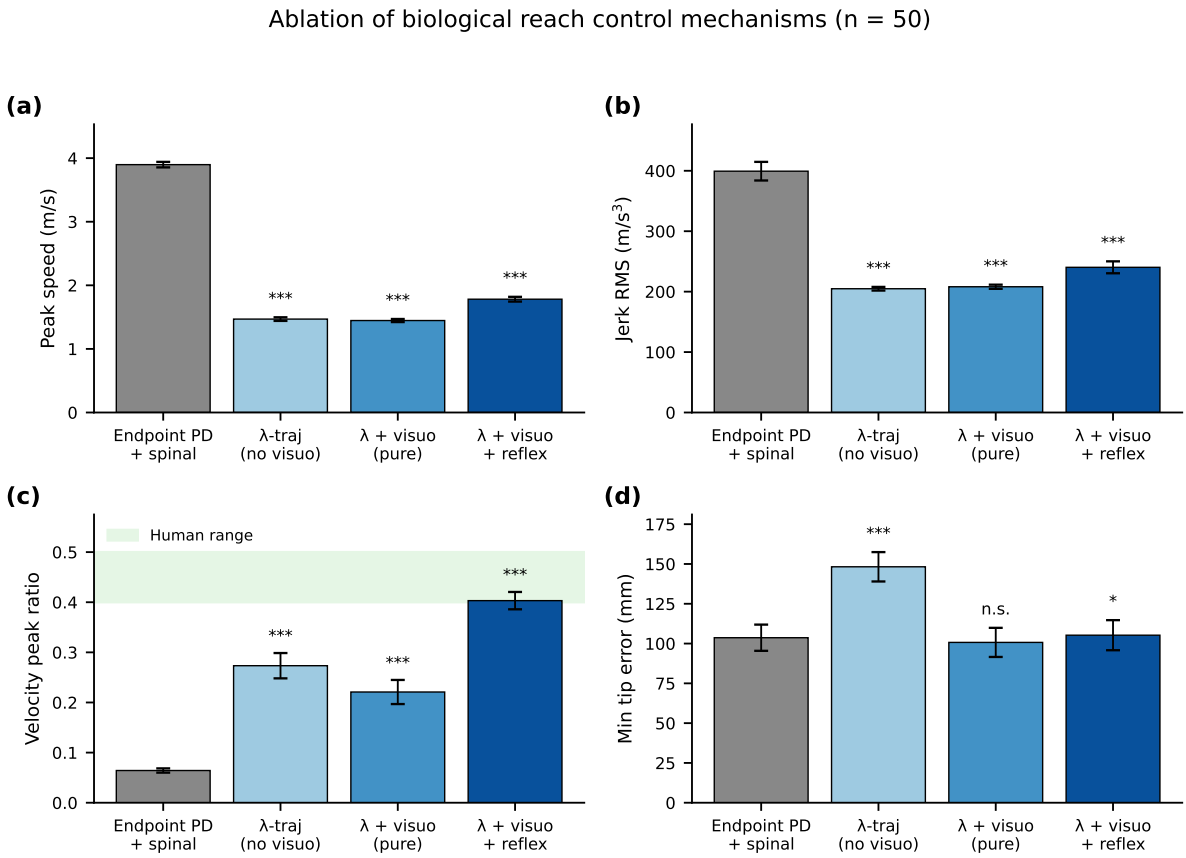

Fig. 1 — Ablation of biological reach control mechanisms. Minimum-tip-error distributions overlap across all controllers (panel d), but only the biologically-grounded λ-EP variants partially reproduce the peak-timing and smoothness invariants of human reach. **(a)** Peak tip speed. Endpoint-PD + spinal (gray) overshoots at ≈ 3.9 m s^{-1}, whereas the static λ-trajectory baseline (light blue) and the λ + visuomotor controller with reflexes (dark blue) remain within ≈ 1.5–1.8 m s^{-1}, matching reported human values for ~ 50 cm reaches (Morasso 1981). **(b)** Root-mean-square jerk. Reflexes do not increase smoothness costs. **(c)** Velocity-peak ratio (vpr): the time of peak speed normalised by movement duration. Only the +reflex controller falls inside the human band (0.40–0.50, green). **(d)** Minimum tip-error. The endpoint-PD + spinal, pure λ + visuomotor, and +reflex headline conditions cluster near ≈ 100 mm and remain within the pre-defined ± 20 mm practical equivalence margin (§2.7, §3.1); the no-visuomotor λ-trajectory condition reaches less accurately (≈ 148 mm), consistent with §3.2 where visuomotor feedback is identified as the primary accuracy component. Bars show mean ± SEM ($n = 50$ randomised seeds per condition). Significance is reported relative to the endpoint-PD + spinal baseline (§2.6) by the seed-paired Wilcoxon signed-rank test (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$; n.s. = not significant), in line with the paired seed design described in §2.7.

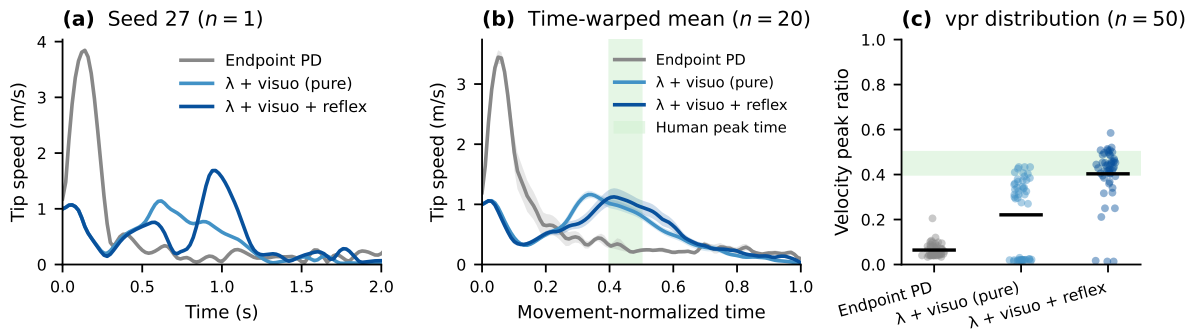

Fig. 2 — Velocity profiles. Adding spinal reflexes to the λ-EP controller transforms a multi-peak speed profile into a single dominant profile whose peak timing falls inside the human reference band. **(a)** Tip-speed profiles for a single representative seed (seed 27, $n = 1$). The endpoint-PD + spinal baseline (gray) produces an initial ballistic spike followed by oscillatory residuals. The pure λ + visuomotor controller (light blue) shows two prolonged peaks. Adding stretch reflexes (dark blue) yields a single dominant peak near 1.7 m s^{-1} with peak timing inside the human band; the profile retains a small initial transient and a residual right- skewness (see §3.1). **(b)** Time-warped mean speed across $n = 20$ seeds (the ablation seed list of §3.2; movement window defined by speed > 0.02 m s^{-1}). Shaded band $= \pm 1$ SEM. Green band marks the human peak-time range (0.40–0.50 of normalised movement time). The +reflex condition (dark blue) is the only one whose mean profile peaks inside this band. **(c)** Per-seed velocity-peak ratio across the headline $n = 50$ seeds of §3.1. Black bars = mean. Green band = human reference range (Morasso 1981; Flash and Hogan 1985). The change of n between panels (b) and (c) is intentional: panel (b) shares the $n = 20$ seed list with the ablation analysis (Fig. 4), and panel (c) reports the headline $n = 50$ distribution.

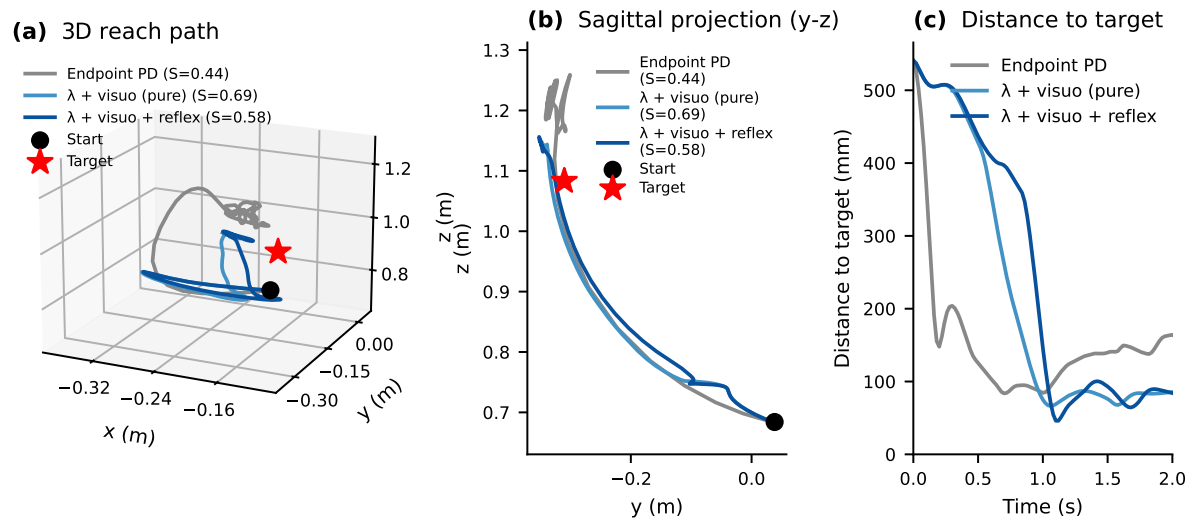

Fig. 3 — End-effector trajectories. The λ-EP controller produces straighter and more direct end-effector paths than the engineering baseline. **(a)** 3-D reach paths for a representative seed (seed 27). Black circle = start, red star = target. The straightness ratio S (straight-line distance / path length, higher = straighter) is

shown in each legend entry. **(b)** Sagittal projection (y–z plane) of the same trajectories, illustrating the qualitative difference between the early ballistic overshoot of endpoint-PD and the smoother arc of the λ-EP controllers. **(c)** Distance-to-target time courses. The λ-EP controllers approach the target monotonically; endpoint-PD oscillates around the residual ≈ 100 mm offset throughout the 2 s movement window.

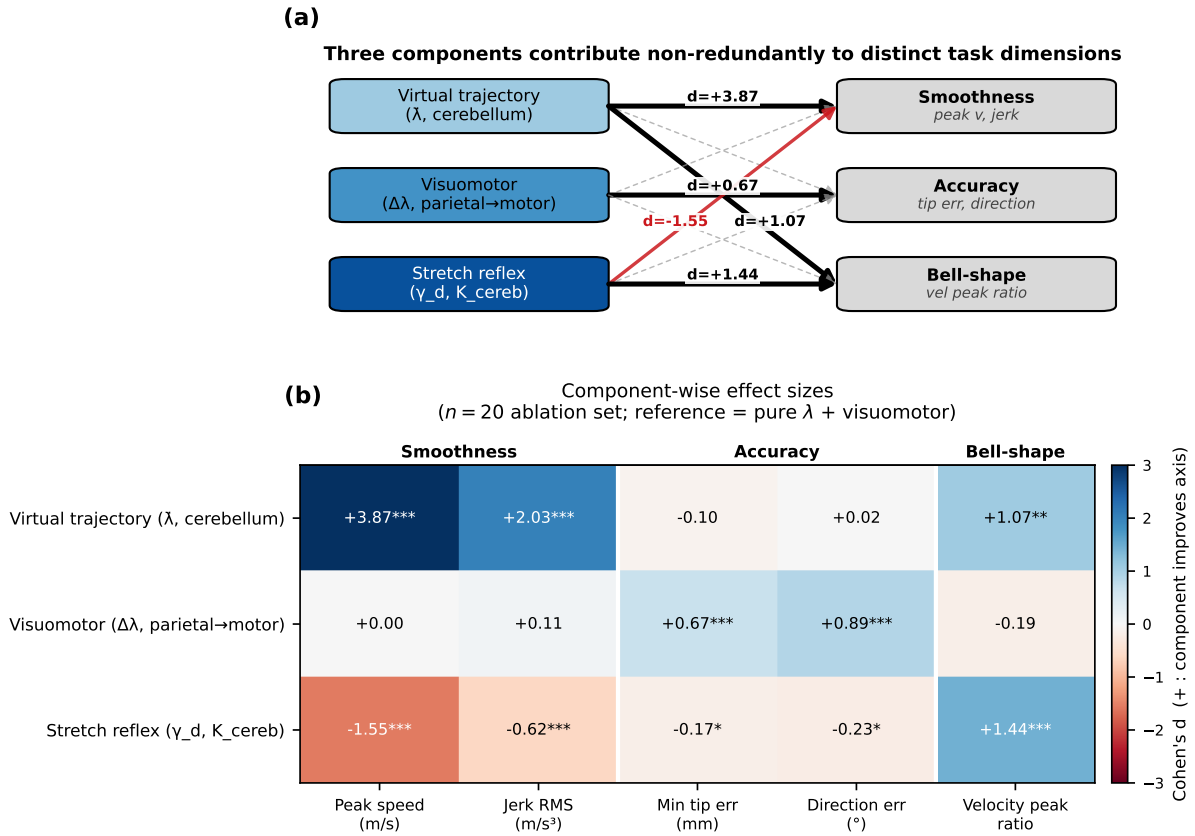

Fig. 4 — Component contributions of the three controller layers. Each controller component contributes preferentially to one task dimension (smoothness, accuracy, or velocity-peak timing), with two practically-sized off-diagonal effects. **Reference configuration is the pure λ + visuomotor controller (no reflexes, no cerebellar correction), not the endpoint-PD + spinal baseline of §3.1; signed Cohen's d is reported with the convention positive = improvement (lower for error / peak speed / jerk; higher for vpr). (a)** Component-to-axis schematic. Three components — virtual trajectory ($\dot{\boldsymbol{\lambda}}$, light blue), visuomotor feedback ($\Delta\boldsymbol{\lambda}$, mid blue), and the spinal reflex layer (dark blue; the optional cerebellar correction is treated separately in §3.3 and is not part of this ablation) — project onto three behavioural axes (smoothness, accuracy, velocity-peak timing). Black solid arrows mark improvements with $|d| > 0.5$ (signed Cohen's d based on the primary metric for each axis: peak speed, minimum tip error, and velocity-peak ratio respectively); red arrows mark degradations of equivalent magnitude; thin dashed gray arrows mark weak effects ($|d| \leq 0.5$). **(b)** Signed Cohen's d for each component × metric pair ($n = 20$ seeds per condition; data from the F13 ablation). Positive values (blue) indicate that adding the component pushes the met-

ric in the behaviourally desirable direction (lower for error / speed / jerk; higher for vpr). Asterisks denote the seed-paired Wilcoxon signed-rank test (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$); Cohen's d is shown as the distribution-level effect size. The block-diagonal pattern demonstrates that the three components contribute non-redundantly, with two practically-sized off-diagonal terms — virtual trajectory has a secondary positive effect on bell shape ($d = +1.07$**) and reflexes have a moderate negative effect on peak-speed control ($d = -1.55$***) — and two small but paired-significant accuracy costs of reflexes ($d = -0.17$* on minimum tip error, $d = -0.23$* on direction error).

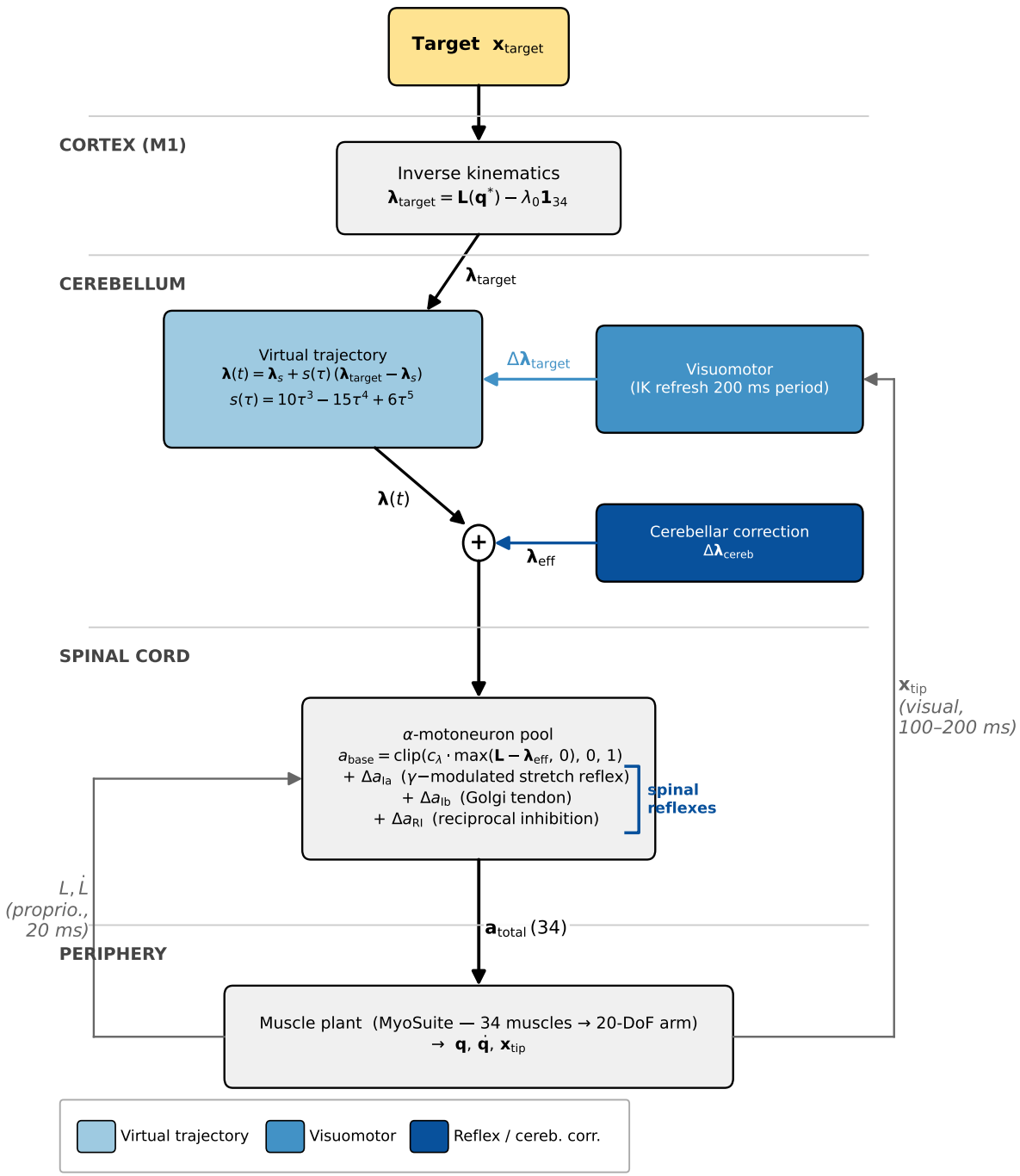

Fig. 5 — Controller architecture. Block diagram of the neuromusculoskeletal

reach controller, organised by anatomical level. Cortical, cerebellar and spinal blocks are kept distinct; the figure is a functional schematic, not a strict neuroanatomical mapping. A target position $\mathbf{x}_{\text{target}}$ provided by the visual system (top) drives an inverse-kinematics solver at the **cortical/planning** level that returns the equilibrium muscle thresholds $\boldsymbol{\lambda}_{\text{target}} = \mathbf{L}(\mathbf{q}^*) - \lambda_0 \mathbf{1}_{34}$ and generates a minimum-jerk virtual trajectory $\boldsymbol{\lambda}(t) = \boldsymbol{\lambda}_s + s(\tau)(\boldsymbol{\lambda}_{\text{target}} - \boldsymbol{\lambda}_s)$, $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ (Flash and Hogan 1985). A separate **parietal–motor visuomotor block** updates $\boldsymbol{\lambda}_{\text{target}}$ from the observed tip position every 200 ms, and an optional **cerebellar branch** (dashed in the figure) adds a predictive correction $\Delta\boldsymbol{\lambda}_{\text{cereb}}$ from a CfC forward-model prediction error (this branch is the negative-result configuration of §2.5 and §3.3 — disabled in the headline controller). The resulting effective threshold $\boldsymbol{\lambda}_{\text{eff}}$ projects to the **spinal** α-motoneuron pool, where a base activation $\mathbf{a}_{\text{base}} = \text{clip}(c_\lambda \cdot \max(\mathbf{L} - \boldsymbol{\lambda}_{\text{eff}}, 0), 0, 1)$ is augmented by an Ia stretch reflex $\Delta\mathbf{a}_{\text{Ia}}$ (γ-MN-modulation slot retained but disabled, see §2.4), a Golgi- tendon component $\Delta\mathbf{a}_{\text{Ib}}$, and reciprocal inhibition $\Delta\mathbf{a}_{\text{RI}}$. The summed activation $\mathbf{a}_{\text{total}}$ drives the MyoSuite myoArm muscle plant (34 muscles, 20-DoF arm), whose output state $(\mathbf{q}, \dot{\mathbf{q}}, \mathbf{x}_{\text{tip}})$ feeds back to the spinal pool with a 20 ms proprioceptive delay (left loop) and to the visuomotor block with a 100–200 ms visual delay (right loop). Component colours match those used in Figs 1–4.

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